An Instance-Based Multitask Graph Network for Complex Facility Recognition in Remote Sensing Imagery

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***Abstract*— With the availability of very high-resolution remote sensing imagery, the fine-grained recognition of complex geospatial facilities has become possible. We can view these facilities as a combination of component objects with specific functions and distribution. However, the existing methods are insufficient in modeling spatial relations of component objects. In this article, we propose an instance-based multitask graph network (IBMG-Net) for complex facility recognition. Specifically, we perform pixel-level component objects prediction and facility recognition simultaneously and achieve performance improvement of both tasks by joint multitasking training. Given the component information, we build an instance-based graph neural network (IBGN) where components are defined as nodes and their spatial relations are encoded as edges. The IBGN module aims to flexibly model spatial relations of complex facility. To enhance the feature representation of component objects, we utilize the multiscale region of interest module (MS-ROI) to retain all scale-specific features and the sparse context information module (SCM) to aggregate long-range context information. In addition, we build a new multitask dataset for complex facility recognition in remote sensing (MCF dataset) to verify the effectiveness of our method and alleviate the lack of pixel-level labeled multitask datasets in remote sensing. Extensive experiments on MCF also indicate that the significant performance improvement of our approach to complex facility recognition.**

***Index Terms*— Complex facility, component information, contextual information, graph neural network (GNN), instance segmentation, remote sensing imagery, scene recognition.**

I. Introduction

COMPLEX geospatial facilities, such as sewage treatment plants and nuclear power plants, are often presented as specific functions and components distribution.

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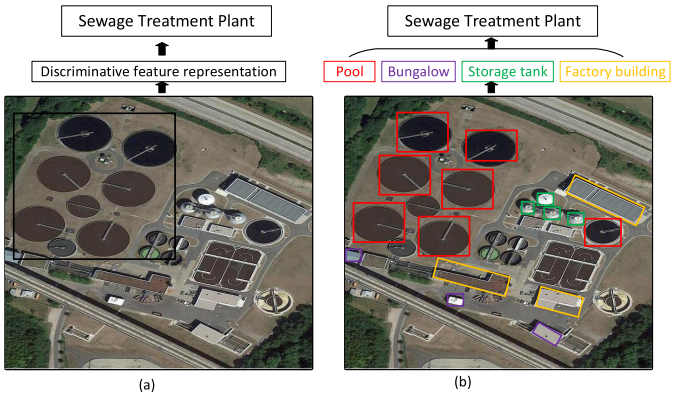


Fig. 1. Comparison of different methods for complex facility recognition. (a) Feature-based methods attempt to obtain a discriminative feature representation, which ignores the geometry and spatial interactions of the components. (b) Component-based methods decompose the scenes into a series of component objects (such as factory buildings, storage tank, pool, and so on), and model spatial relations between these objects.

The fine-grained recognition of complex facilities has crucial implications in urban planning, environmental monitoring, land use/land cover (LULC) classification, etc [1]–[5]. Unlike the traditional scene recognition task, complex facility recognition requires identifying component objects and their spatial relationships [6]–[8]. It is reasonable to decompose the complex facility into several fixed component objects, such as columnar buildings, factory buildings, pools, etc. Note that the component objects in different facilities vary significantly in the category and spatial distribution. We can recognize various complex facilities by the discriminative component representation, which is also named as the component-based method in the Fig. 1(b).

However, fine-grained recognition of complex facility is challenging due to some characteristics of remote sensing imagery. Remote sensing scenes always contain some local regions irrelevant to the label, which we call the background [9]. The background can cause significant interference in poor lighting conditions and dense spatial distribution. At the same time, component objects in complex geospatial facilities occur in high density, huge scale variation, arbitrary shapes, and orientation [10]–[12]. Most existing methods attempt to learn discriminative feature representation for performing scene recognition [13]–[15], as shown in Fig. 1(a).

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However, such semantic representations are not enough to represent the distinctive shapes, sizes, and spatial relationships.

To tackle these shortcomings of feature-based methods, Vatsavai *et al.* [6] propose a bag of words (BOW) model to perform complex facility recognition based on the Latent Dirchlet Allocation method. However, it neither considers the spatial interactions among components nor the geometry of the component objects. Inspired by the robust feature representation of CNN, Gadiraju *et al.* [7] evaluate the performance of the CNN models in classifying complex facilities. As shown in Fig. 2(a), CNN-based methods are the most widely used framework in scene recognition, which can extract discriminative features and generate robust semantic representation [2], [16], [17]. However, several shortcomings remain in the current CNN-based works for complex facility recognition: 1) the existing methods use CNN as a feature extractor but fail to model the spatial relations in complex facility; 2) the feature extraction module is not flexible enough to handle scale-variant component objects; and 3) the background information can be quite strong and it is difficult to capture the confusing component objects.

Recently, the attempts to explore component objects and their spatial relations have been a research hotspot in scene recognition [18]–[21]. Considering the location and scale of various objects, Wang *et al.* [22] try to localize discriminative local regions by pretrained region proposal model. Although these regions are densely sampled in images and located in a grid, the fixed shapes still constrain the network's ability to capture the shape-variant components. At the same time, some earlier works [23]–[25] learn several prototypes for each scene. Such methods are not comprehensive enough to cover the large structural diversity of different scenes. Inspired by the application of GNN in image-understanding tasks [26], [27], Chen *et al. [28]* proposes layout graph network (LGN), which transforms the region proposal information and spatial relations into a discriminative scene representation.

However, these works ignore the semantic information of components and make it difficult to recognize complex facilities. In contrast, our motivation is to design a framework to extract component information and model spatial relations more effectively. As a result, we perform component instance segmentation and build a novel graph neural network (GNN) for recognizing complex facility. As shown in Fig. 2(b), instance segmentation methods can provide pixel-level semantic information in addition to distinguishing different components. Thus, it is possible to take full advantage of location and semantic information for each component.

In this article, we propose instance-based multitask graph network (IBMG-Net), a novel multitask framework, which performs pixel-level component extraction (instance segmentation) and complex facility label prediction (scene recognition) in remote sensing imagery. As illustrated in Fig. 2(c), considering the ability of the graph to preserve diverse spatial relations, we construct instance-based graph neural network (IBGN) module with the semantics and locations of component objects. More specifically, we define the instance features as nodes and encode spatial relations between nodes as edges. Through the graph convolution (GC) and post-process operations, the instance information and spatial relations can be transformed into a discriminative scene representation. Besides, in order to handle the objects with variant scale and suppress the strong background information, IBMG-Net retains multiscale features and proposes sparse context information module (SCM) to perform further feature extraction. Moreover, inspired by the multitask learning, IBMG-Net achieves performance improvement of instance segmentation and scene classification by joint multitasking training.

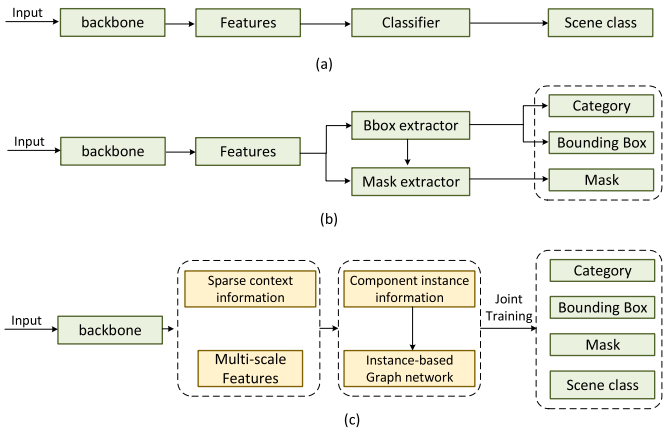


Fig. 2. Illustration of the different pipeline. (a) General framework of CNN-based scene recognition methods. (b) General framework in instance segmentation. It extends two-stage object detection methods by adding a branch for predicting masks. (c) IBMG-Net is a multitask framework. It uses sparse context information and multiscale features to enable the network to focus on the key components in instance segmentation. And it uses an instance-based graph network to model spatial relations and complete scene recognition.

In general, the contribution of this article can be summarized as follows.

1) We propose a novel multitask framework, IBMG-Net, for complex facility recognition in remote sensing imagery. Specifically, the framework can perform pixel-level prediction of component objects along with recognition of combinatorial geospatial facilities. As a result, we achieve performance improvement for both tasks by joint training.
2) Considering the geometry, semantic information, and spatial interaction between the various components, we extract the component instance information and build an IBGN to model the spatial relations. It is the first attempt to construct a GNN through instance information in remote sensing imagery.
3) To mitigate the interference of variant object scale and strong background information, we propose the multiscale region of interest module (MS-ROI) and SCM in the feature extraction stage. MS-ROI retains all scale-specific features, and SCM focuses on several representative positions for stressing the key component objects.

Meanwhile, to verify the effectiveness of our method, we build a new multitask dataset of complex combinatorial facilities (MCF dataset). Compared with other datasets in remote sensing, our dataset exhibits the following distinctive characteristics: 1) high spatial resolution; 2) precise pixel-level

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instance annotations; and 3) 12 highly representative component categories, as well as 8 typical complex geospatial facility categories. The dataset will be published later. The remainder of the article is organized as follows. In Section II, we briefly introduce some related methods. Then, we explain the details of network in Section III and show the experiment results in Section IV. Finally, Section V concludes the article with result discussion.

II. Related Work

A. Feature-Based Scene Recognition

In the early works, handcrafted features-based methods (such as SIFT [29]) are the mainstream methods in scene recognition. Based on handcrafted features, bag-of-features methods (such as VLAD [30] and Fisher Vector [31]) achieve great performance improvement on scene recognition, which have also served as a good inspiration for many works [32]–[34]. However, these methods exploit local features in an orderless manner, which ignores the relation between local features and global semantic representation.

With the rapid development of convolutional neural networks in vision understanding tasks, CNN-based methods are widely utilized and outperform the handcrafted features methods on scene recognition [4], [5], [14], [28]. Most existing approaches use pretrained models from large dataset (like ImageNet) and fine tune on a remote sensing dataset [9]. Specifically, some works use the end-to-end CNN framework to extract discriminative features and obtain a robust global semantic representation [35], [36]. And using classic CNN models (e.g., AlexNet, VGG, ResNet) as the feature extractor is another trick for scene recognition in remote sensing [37]. However, CNN-based methods tend to capture global semantics but always suffer from the subtle difference among classes.

Recently, the CNN-based methods in remote sensing scene recognition usually design new architecture to extract scene features [1], [2], [4], [5]. Some existing works combine CNN models with bag-of-feature methods or dictionary-based methods [13], [38] to enhance discriminative local information. López-Cifuentes *et al.* [39] propose a context information module with the idea of semantic segmentation to enhance the image feature maps. To improve the semantic representation ability of RS-oriented scene categories, Li *et al.* [3] construct a new remote sensing knowledge graph (RSKG) and propose to generate a Semantic Representation of scene categories by representation learning of RSKG.

However, these works either encounter the problem of noise regions caused by predefined grids, or insufficient of the spatial structural information.

B. Component-Based Scene Recognition

Consider the spatial relations of complex facility in remote sensing imagery: the use of the component object information can greatly influence the performance of the scene recognition models. And we refer to these works as component-based methods. Bi *et al.* [9] highlight the local semantics relevant to the scene label and transform remote sensing scene recognition into multiple-instance learning (MIL) problem. However, the MIL-based methods suffer from the lack of the component object annotations, which leads to poor interpretability in their semantic representation.

Significantly, most existing works try to localize the component objects by pretrained object detection models. Wang *et al.* [22] investigate a novel framework to obtain region proposals and encode features of these region proposals to discriminative scene representation. Consider the weakness of component semantic information: we perform instance segmentation to capture the semantic information of diverse component objects.

Instance segmentation is one of the research hotspots in computer vision. Unlike object detection task and semantic segmentation task, instant segmentation extracts individual instances from the original image and distinguishes them from each other. He *et al.* [40] add a mask prediction branch to the faster RCNN [41] and propose the ROI Align module to achieve better performance. Cai and Vasconcelos [42] extend the cascade R-CNN architecture to the instance segmentation, where the output of each stage fed into the next one for higher quality refinement. And Chen *et al.* [43] improve the information flow by incorporating cascade and multitasking at each stage in [42].

Besides, it is critical to model the spatial relations of the component objects in scenes. Some early works [18]–[21] explore spatial relations by learning a fixed number of structures for each scene category, where we also name the learned structures as the scene prototypes. However, the scene prototypes are not comprehensive enough to cover the large structural diversity of different scenes.

C. Graph Neural Network

GNNs are the methods that can efficiently learn the spatial relationships of graph-structured data [44]. GC (GC) network [45] and graph attention network (GAT) [46] generate node weights through the interactions of degree matrics and update node information using feature information of one-hop nodes. GraphSAGE [47] aggregates the information of neighboring vertices based on the idea of maximum pooling, while graph isomorphic network (GIN) [48] sums the node embeddings.

Recently, some researchers apply GNN in computer vision tasks. Chen *et al.* [28] propose a LGN for scene recognition and incorporate two topological structures (spatial and semantic) through GC. Some works try to apply GC into the semantic segmentation task and model the long-range context relations [26], [49]. Zhang *et al.* [50] model an efficient and yet flexible nonlocal relation representation based on a novel GNN framework. Yu *et al.* [51] build a representative graph (RepGraph) layer to dynamically sample related features to compute the response of one node, which greatly reduce the computational redundancy caused by nonlocal networks. Luo *et al.* [52] construct a cascade graph model to exploit multiscale, cross-modality information for salient object detection. Zhai *et al.* [27] design a novel mutual graph learning model to handle different complementary relations

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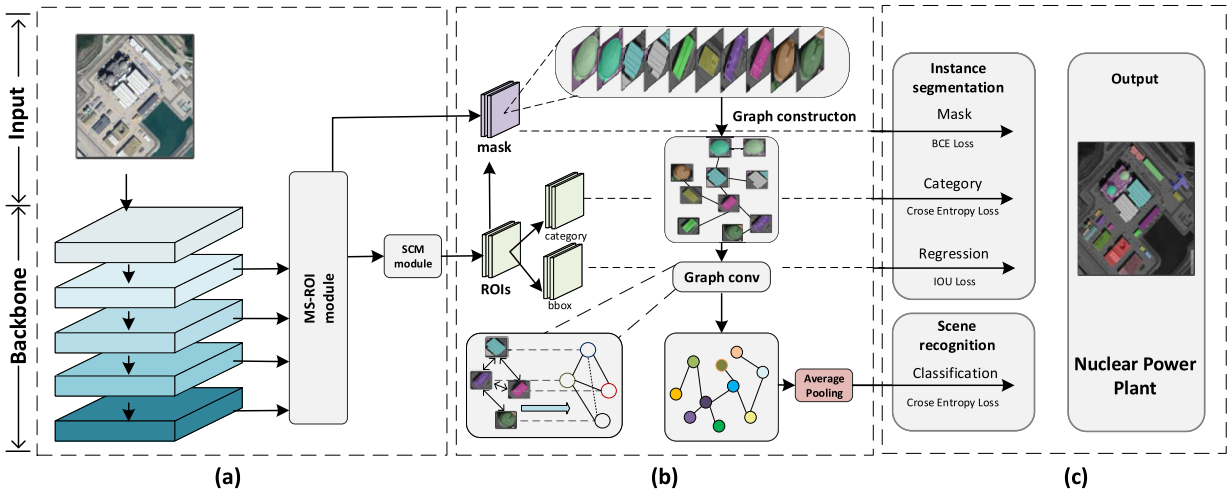


Fig. 3. Framework of our network. It consists of three modules: (a) Multi-scale Feature Extraction (MSFE); (b) Instance-based Graph Network; (c) Multi-task Interpretation (MTI). The MSFE module extracts multiscale features and aggregates sparse context information. The IBGN module constructs a GNN based on the instance information to aggregate the spatial structure information of the scene. And the MTI module jointly trains the two tasks to achieve model performance improvement.

to maximize information interactions. Liang *et al.* [53] detect ground objects in remote sensing imagery and construct GNN to learn spatial location features. As a result, these attempts provide a new idea for complex facility scene recognition in remote sensing imagery.

To stress the component information while suppressing the strong background, It's crucial to model long-range context information in understanding tasks. Attention mechanism based on the idea of nonlocal networks [54], [55] mostly try to correlate the output response of a location with other locations. However, since the nonlocal network [56] needs to derive an affinity matrix by modeling the importance of each position concerning other pixels, we can regard the nonlocal networks as a densely connected GNN with each pixel as nodes. The affinity matrix is equivalent to the weight matrix of the edges in the GNN. Therefore, we can use the GNN instead of the traditional global attention mechanism to model long-range context information.

III. Methodology

A. Problem Formulation

We formulate the complex facility recognition task in remote sensing imagry as a combination of two-stage. First, we perform instance segmentation and complete the pixel-wise prediction of component objects. Second, we build a GNN based on the instance information and aggregate spatial relations using a GC operation. Besides, we predict facility categories according to the discriminative component representation.

As shown in Fig. 3, our network consists of three main stages: multiscale feature extraction (MSFE), IBGN, and multitask interpretation (MTI). The MSFE consists of two modules: SCM and MS-ROI.

SCM fuses the long-range context information in the image, which is inspired by the Representative Graph layer in [51] This module only focuses on a sparse set of representative locations and is less computationally intensive compared with the nonlocal neural network. At the same time, MS-ROI retains features with various scales and adapts to complex facilities with huge scale differences. In contrast, the traditional ROI extraction methods select single-layer feature map information according to specific rules [57].

IBGN is a novel GNN framework. Here we define the predicted instance information (aforementioned component objects) as nodes, the features of the instances as node features, and the spatial distances between each node as edges. By using GC, we can transform the instance information and spatial relations between instances into a discriminative image representation for complex facility recognition.

MTI uses a multitask loss function to train the two tasks jointly. Consider complex facility recognition shares complementary information with the instance segmentation: joint multitasking training is equivalent to each one task acting as a regularizer for others. Thus, we can achieve performance improvement of these two tasks by MTI.

Due to the robust feature extraction capability of the instance segmentation network and the flexible spatial information modeling capability of the GNN, IBMG-Net can effectively utilize multiscale semantic information and achieve performance improvement of two tasks by joint multitasking training.

B. Sparse Context Information Module

To aggregate long-range context information, a series of attention mechanisms based on nonlocal networks have been proposed. The nonlocal network [56] is equivalent to a global densely connected graph, which can be formulated as

$$\tilde{x}_i = \sum_{\forall j} f\left(x_i, x_j\right) g\left(x_j\right) \tag{1}$$

where x is the input feature map, x_i is an input position, and x_j is the index that enumerates all possible positions. $g(x_j) = W_g x_j$, W_g is a weight matrix to be learned. The function f is the function that computes the affinity between

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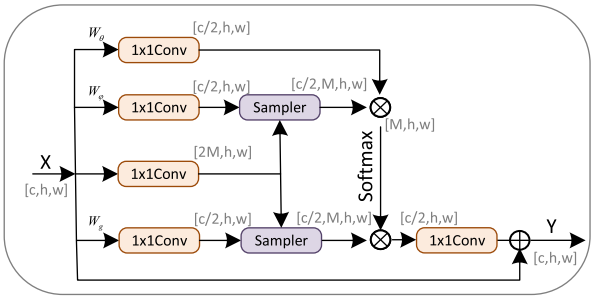


Fig. 4. SCM. X is the feature map that is shown as the shape of the tensor, e.g., $c \times h \times w$, where c, h, and w represent channels, height, and width, respectively. M is the number of representative nodes.

i and all j. The function g computes the representation of input x_i at all position j.

Here f can be defined as a dot-product similarity

$$f(x_i, x_j) = \delta\left(\theta(x_i)\phi(x_j)^T\right) \tag{2}$$

where $\theta(x_i) = W_\theta x_i$ and $\phi(x_j) = W_\phi x_j$ are two embeddings, W_θ and W_ϕ are two weight matrix, δ is the softmax function.

However, there are potential relations among features, but each particular location may only be closely related to a limited number of other locations. To model such context relations, inspired in [51] and [58], we don't need to calculate the response with all potential positions. Instead, we use a SCM based on nonlocal network to sample a limited number of representative nodes. To identify these representative nodes, we add convolution operation to regress an offset matrix with 2M channels. With the offset matrix, the sampler will calculate the positions of M representative nodes.

As shown in Fig. 4, the X is the current feature map. We use a 1 * 1 convolution operation to obtain an offset matrix so as to sample M representative nodes. The operation of the sampler can be formulated as follows:

$$F(x_i) = \left\{x_{i+\Delta p_n} | n = 1, 2, \ldots, M\right\} \tag{3}$$

where Δp_n is offset coordinates of M representative nodes, and n is the certain position of M representative nodes.

Considering that the learned offsets in the above coordinate regression are often fractional, we use the bilinear interpolation to compute the feature of the fractional position through the features of four integer adjacency positions

$$x_{i+\Delta p_n} = \sum_{\forall t} G(t, p_n) x_t \tag{4}$$

where t is the position of four integer adjacency pixel and G is bilinear interpolation kernel.

With the sampling function $F(x(i))$, we are able to obtain M representative node features $x(p_i)$, and then aggregate the associated features at each location

$$\tilde{x}_i = \sum_{\forall n} \delta\left(\theta(x_i)\phi(F(x_i))^T\right) g(F(x_i)) \tag{5}$$

where θ, ϕ, and g are three embeddings, which are the same as these in the nonlocal network mentioned above.

In general, the overall structure can be expressed by the following:

$$y_i = W\tilde{x}_i + x_i \tag{6}$$

where $\tilde{x}_i$ is the node feature after aggregating information of M representative nodes, W is the weight matrix, which should be initialized to zero.

C. Multiscale Region of Interest Module

The traditional region proposal networks choose a particular layer in feature pyramid network (FPN) by the width and height of the ROI [57]

$$k = \left\lfloor k_0 + \log_2\left(\sqrt{wh}/224\right)\right\rfloor \tag{7}$$

where k_0 is the highest level feature map and 224 is the typical image size.

However, the selection of the single layer of FPN limits the networks's description [59]. Instead, our MS-ROI use multilayer feature maps in FPN. As shown in Fig. 5, by separate ROI Align operation in preprocessing phase, we obtain multi fixed-size output in different layers of FPN. By performing comparison of different feature aggregation operations, such as concatenation, summation, and gating mechanism in the dynamic FPN [60], we aggregate multi outputs by features summation that allows the network to reach a convergence state more easily. We will elaborate on different operation in Section IV. In summary, our methods can be formulated as follows:

$$f(x) = \left\{\sum_{\forall k} (\text{roi}(x_k)) | k \in \{P_3, P_4, P_5\}\right\} \tag{8}$$

where $\text{roi}(x_k)$ is the operation of ROI Align, $\{P_3, P_4, P_5\}$ is the multilayer feature activations output in ResNet.

To enhance the power of the network to learn global semantics, we append an extra step as the postprocessing. The existing works have proven that adding nonlocal network and other attention mechanisms has the best performance. Instead, we use the SCM module for postprocessing. In the Section III, we show that the SCM can realize the aggregation of long-range context information more effectively than the nonlocal network.

D. Instance-Based Graph Neural Network Module

The IBGN module is a novel structure, which utilizes the location and semantic information obtained by the instance segmentation stage. The module mainly includes two processes: graph network construction and GC update. We define the component instances as nodes and the spatial distances between each node as edges. Thus, we get the node set $V = \{v_1, v_2, \ldots, v_n\}$ and the edge set $E = \{e_1, e_2, \ldots, e_m\}$. As long as we get the instance information in remote sensing imagery, we can construct such GNN $G = \{V, E\}$.

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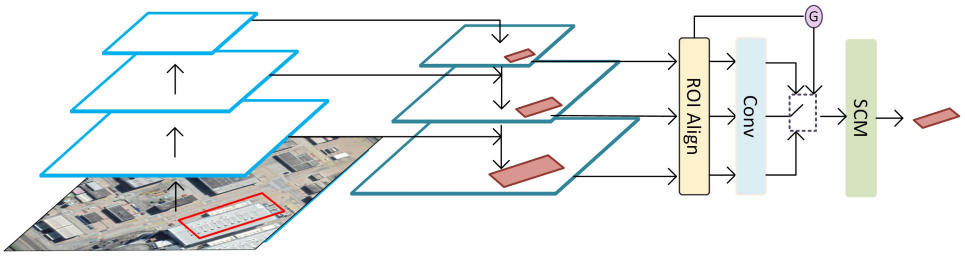


Fig. 5. MS-ROI. The module uses ROI Align and convolution operations to obtain a fixed size output, aggregate features by the summation operation, and uses an SCM to complete the postprocessing.

1) Graph Network Construction: The instance information, such as the location and semantic information, is the key to mining the spatial structure of the combination facilities in remote sensing imagery. Specifically, we use the location information of the bounding box, $\text{bbox}_i = [y_1, x_1, y_2, x_2]$, to model the spatial relations. Inspired by [28], we normalize the location information of each instance as follows, and generate the edge set E and adjacency matrix A:

$$d_i = \left[\frac{y_1}{w}, \frac{x_1}{h}, \frac{y_2}{w}, \frac{x_2}{h}\right] \tag{9}$$

$$e_{ij} = \|d_i - d_j\| \tag{10}$$

$$a_{ij} = \frac{\exp(e_{ij})}{\sum_{i \neq j} \exp(e_{ij})} \tag{11}$$

where w and h are the width and height of the remote sensing imagery, e_{ij} is the edge between instance i and j, and a_{ij} is the element of the adjacency matrix A.

To combine the instance features and predicted semantic labels by instance segmentation, we can use the prediction results of the instance categories, as well as using instance features. These attempts enable our GNN to utilize the feature and semantic information of component objects.

2) Graph Convolution Update: After the GNN construction, we get a graph $G = \{V, A\}$. Considering the superior performance of GCN on graph-structured data, we use GC to aggregate the node information. And we construct the degree matrix D and Laplacian matrix L based on the adjacency matrix as follows:

$$D = \text{diag}(d_1, d_2, \ldots, d_n), \quad d_i = \sum_j A_{ij} \tag{12}$$

$$L = I - D^{-\frac{1}{2}} \text{AD}^{-\frac{1}{2}}. \tag{13}$$

GC is formulated as follows. Here we also construct a weight matrix W to be learned and combine the Laplacian matrix L and the adjacency matrix A to update the node features

$$Z = \sigma \text{LVW} = \sigma\left(\left(I - D^{-\frac{1}{2}} A D^{-\frac{1}{2}}\right) V\right) W \tag{14}$$

where σ is the nonlinear activate function (here we use ReLU). The parameters of W are learned by gradient descent and initialized randomly, and the Z is the output.

Based on the above discussion, we are able to divide the GC process into two stages, as shown in Fig. 6. First, the Laplace smoothing operation aggregates node features and then learns the weights of edges through the adjacency matrix,

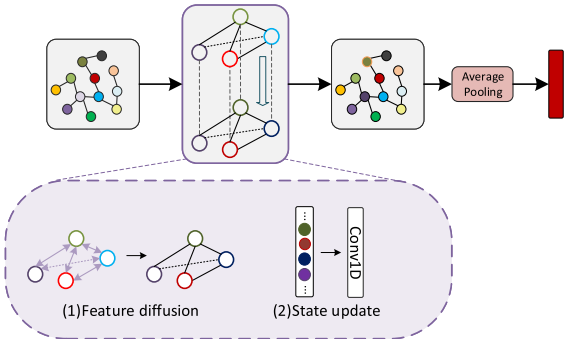


Fig. 6. IBGN. We define component instances as nodes, the instance's feature information as node's feature, and the spatial distances between each node as edges. And the GC operation can be viewed as two stages below: feature diffusion and state update.

which reflects the global features associated with each node. After feature diffusion, each node has received all necessary information and its state can be updated through a linear transformation.

The GC also has stronger interpretability compared to the convolution operation. Suppose two nodes contain features focused on columnar buildings and factory buildings. In that case, learning the strong connection between these two nodes can strengthen the discriminative representation for the downstream task of recognizing the facility areas. Thus, after the aggregation of node-by-node and channel-by-channel features, each instance can get richer spatial information to guide the subsequent tasks.

E. Multitask Interpretation Module

We define the loss function for each subtask separately as follows:

$$L_{\text{bbox}}(c_i, r_i, \hat{c}_i, \hat{r}_i) = L_{\text{cate}}(c_i, \hat{c}_i) + L_{\text{reg}}(r_i, \hat{r}_i) \tag{15}$$

$$L_{\text{mask}}(m_i, \hat{m}_i) = \text{BCE}(m, \hat{m}) \tag{16}$$

$$L_{\text{class}} = \text{CE}(s, \hat{s}) \tag{17}$$

where r is the regression bias of the bounding box, c is the category score, m is the predicted instance mask, s is the predicted scene label, the CE is the cross entropy loss, and the BCE is the binary cross entropy loss.

Considering that the exact bounding box is crucial in a multistage model, we use cross entropy loss for L_{cate} and use

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IOU loss for L_{reg}, as follows:

$$L_{\text{reg}}(r, \hat{r}) = -\ln\left(\frac{\text{Intersection}(\text{bbox}(r), \text{bbox}(\hat{r}))}{\text{Union}(\text{bbox}(r), \text{bbox}(\hat{r}))}\right). \quad (18)$$

Considering the complex facility recognition (scene recognition task) and pixel-level prediction of components (instance segmentation task) share complementary information; both the tasks have the potential for improved performance by joint training. The simplest method is to sum the loss functions directly for each subtask, but it can't work well in our task. Inspired by [61], we replace the sum of loss functions with a weighted sum, which brings all losses to approximately the same scale and balances the contributions of each sub-task. At the same time, we use a hyperparameter β to balance the weight of the different tasks. A series of comparison experiments with different value of the hyperparameter β reveal that the overall performance of the network is optimal when β fixed to 0.3

$$L = (1 - \beta)(L_{\text{bbox}} + L_{\text{mask}}) + \beta L_{\text{class}}. \quad (19)$$

IV. Experimental Results

In this section, we present the construction of a new multitask dataset of complex facility in remote sensing imagery (MCF dataset), and the experimental details of the IBMG-Net. These details include evaluation metrics, parameter setting, ablation study, and comparison with other models.

A. MCF Dataset

In the earlier works, Gadiraju *et al.* [7] collected images belonging to four complex facilities: airport, coal plants, gas plants, and nuclear power plant. However, the images in this dataset have poor resolution and lack detailed component annotations. To alleviate the lack of datasets for complex facility recognition task and pixel-level component instance segmentation task, we propose a new dataset with adequate quantities, sufficient instance categories, and precise instance-level annotations. In summary, we build a new well-annotated MCF dataset for the fine-grained recognition of complex facility in remote sensing imagery. The multitask here can be seen as the simultaneous implementation of scene recognition and instance segmentation.

Our MCF dataset has unique challenges because we need to cover a large number of components in every complex facility imagery and give pixel-level annotations. In comparison, most of the existing datasets are annotated with bounding boxes and coarsely localize the object instances. Although several pixel-level labeled datasets [62], [63] for object detection/segmentation in remote sensing have been proposed, only a few of them, such as DOTA [8], iSAID [12], contain multi categories and a large number of instances. Furthermore, these datasets are often limited to a small scale with only a few object categories [10], [62], [63]. Lack of pixel-level instance annotations or scene labels [8], [10], [64], and lack of key component categories (e.g., buildings) [8], [12]. However, the types and distributions of buildings, especially in complex facility of remote sensing scene, tend to relate closely with the scene labels. Therefore, building instances can be seen as vital component category for complex facility recognition. Moreover, due to the lack of building footprints in iSAID and the single instance category in most other datasets, the current datasets are hard to perform fine-grained recognition of complex facility in remote sensing imagery.

To ensure data diversity, we collect a large number of complex combination facilities from the Google Earth images. These images are distributed over several countries, such as China, the United States, Germany, and Japan. Each image has a resolution ranging from 0.3 to 1.2 m, and size ranging from 944×593 to 2632×3056. The pixel-level annotations of the MCF dataset are carried out by an expert group of remote sensing applications, as well as cross-checked and validated with well-defined guidelines, which guarantees the high label accuracy of the dataset.

For the complex facility categories and the component categories in the MCF dataset, as shown in Fig. 7(a), we focus on eight ground combination facilities, i.e., nuclear power plant, fossil-fuel power plant, sewage treatment plant, petrochemical plant, Mining plant, oil field, sports center, and school, that are common artificial facility in remote sensing. We also provide pixel-level annotations for the following twelve component categories in Fig. 7(b), i.e., columnar building, factory building, tall building, bungalow, auxiliary building, court, playground, pool, parking lot, storage tank, artificial bare soil, and other. In summary, components from these categories occur frequently and are important for various real-world applications.

Finally, we construct the dataset that includes 157 images and 4918 instances. In Fig. 8, we analyze the properties of MCF and show statistics of components categories and complex facility categories. As shown in Fig. 8(a), the image count per scene in our dataset is relatively balanced, which usually exists in real world. It is shown that there are some infrequent categories with significantly less components than other categories. For example, bungalow and playground are the most frequent and least frequent categories in complex remote sensing scenes, respectively. Such category imbalance and instance imbalance in Fig. 8(b) and (c) usually exists in both natural and aerial imagery datasets. For dataset split, we use 80% images of the dataset as train set and 20% images as test set. Both images and annotations in our dataset for the dataset will be published soon.

B. Implementation Details

1) Experimental Setup: We use the two-step Mask RCNN as baseline of instance segmentation and ResNet50 as the backbone of our model. At the same time, the resolution of the original images (2000 pixels and above) in the dataset may be too large to be trained, considering the memory limitation. We opt to train and test the dataset on the patches of resize 1024×1024. The model is trained with stochastic gradient descent (SGD) for 40 epochs, and every minibatch

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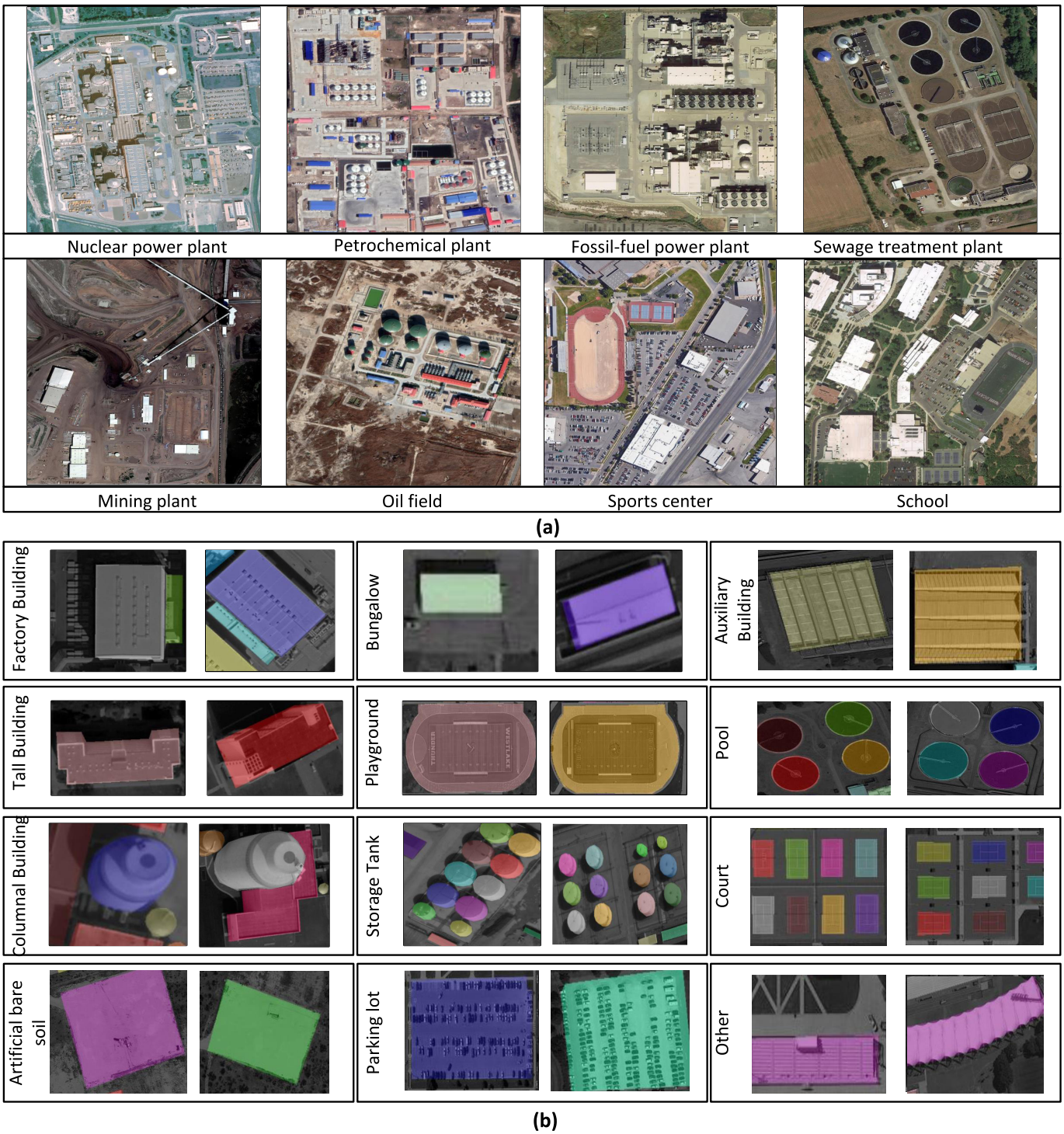


Fig. 7. Samples from the MCF dataset. (a) Eight common artificial complex facilities. (b) Pixel-level annotations for 12 instance categories, and we visualize several samples per each category.

consists of two images. The initial learning rate is set as 0.00125, which is reduced by a factor of 10 in the 30th epoch. Momentum and weight decay are set as 0.9 and 0.0005, respectively. The models are trained on a Tesla V100 with CUDA10.1.

2) Evaluation Metrics: Considering the practical requirements of complex facility scene recognition and instance segmentation, we use the average accuracy (Acc) and the average precision (AP) scores of the COCO metrics [11] as evaluation metrics. Acc refers to the ratio of correctly classified samples to the total samples and is the main evaluation metric in the scene recognition. AP is a common metric to evaluate the performance of object detection and instance segmentation, which can be viewed as the area under the precision–recall curve. Accuracy, precision, and recall metrics can be calculated as follows, where TP, TN, FP, and FN refer

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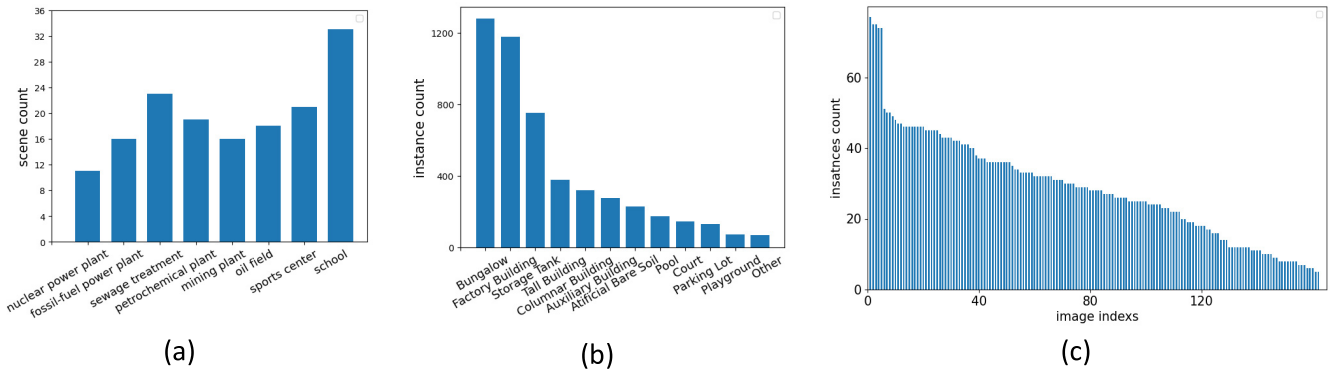


Fig. 8. Statistics of classes and instance of MCF dataset. (a) Histogram of the number of instances per categories (sorted by frequency). (b) Histogram of number of images per scene class. (c) Histogram of number of instances per images.

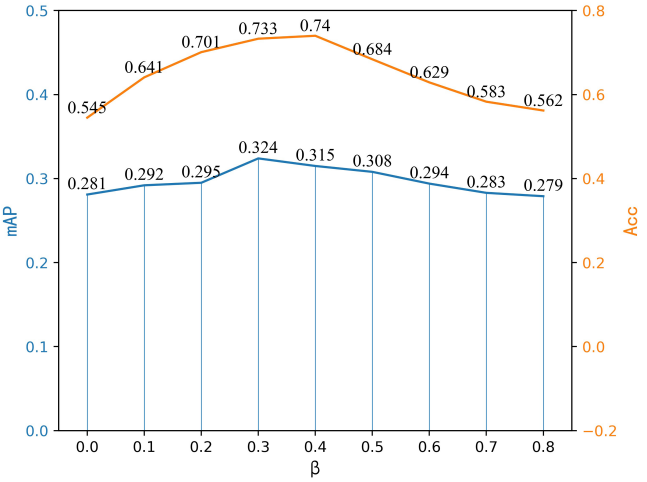


Fig. 9. Impact of the variance of β. We use different value of β for training of baseline and record the results on the test set.

to true positive, true negative, false positive, and false negative, respectively,

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{20}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{21}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{22}$$

In the instance segmentation task, AP is computed with mask IoU, which is the ratio of the overlap and the union between the predicted mask and the ground truth. Therefore, the AP in instance segmentation is also called the mask AP. Furthermore, the standard AP metrics include mAP, AP50, AP_S, AP_M, and AP_L. Specifically, mAP is the mean of the average precision over categories, which is able to measure the performance on all categories. AP50 is the average precision at the IoU of 0.5. And S, M, and L refer to small (10–144 pixels), medium (144–1024 pixels), and large (1024 pixels and above), respectively.

3) Hyperparameter Setting: Our model mainly includes two hyperparameters β (the weights of tasks) and M (the number of sample nodes). To exploit the impact of different values of β, we perform instance segmentation by baseline (Mask RCNN), achieve scene classification by constructing IBGN module, and use different values of hyperparameters β for multitask joint training and record the mAP of the instance segmentation and the Acc of the scene recognition on the test set. As shown in Fig. 9, when β is close to 0, the average accuracy of joint training is comparable to that of single-task training. In contrast, when β is close to 1, joint multitasking training degrades the model performance instead. We find that the scene recognition accuracy is best when the β takes the values of 0.3 and 0.4. To get more accurate instance information, we finally set the β to 0.3.

TABLE I

RESULTS OF DIFFERENT NUMBERS OF SAMPLES IN SCM. WE USE THE RESULTS OF BASELINE (MASK RCNN) AND NONLOCAL NETWORK MODULE AS THE COMPARISON

Model	Sample nodes	mAP	ΔmAP
mask RCNN (baseline) [40]	–	30.20%	–
Non-local Network [56]	–	31.40%	1.20%
SCM	1	31.60%	1.40%
	9	**32.50%**	**2.30%**
	12	31.90%	1.70%
	15	32.10%	1.90%

Table I compares the instance segmentation results of choosing different number of sample nodes. These results show that all the models based on SCMs can achieve performance improvement over baseline (mask RCNN) and nonlocal network. Specifically, the instance segmentation task achieves the best performance when M is 9. And we set 9 as the default in the end.

C. Experimental Results

In this section, we conduct a series of experiments on our dataset and comprehensively evaluate our method on our

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TABLE II
RESULTS OF DIFFERENT METHODS OF COMPLEX FACILITY SCENE RECOGNITION AND INSTANCE SEGMENTATION

Task	Model	mAP	AP50	ΔmAP	Acc
Scene recognition	VGGNet-16 [65]	–	–	–	73.30%
	Resnet50 [65]	–	–	–	78.90%
	Deeper BN-Inception [66]	–	–	–	81.40%
Instance segmentation	mask RCNN (baseline) [40]	30.20%	53.80%	–	–
	mask scoring RCNN [67]	31.40%	54.30%	1.20%	–
	cascade RCNN [42]	33.10%	58.20%	2.90%	–
	Hybird Task Cascade [43]	34.50%	**59.40%**	4.30%	–
Multitask	IBMG-Net(ours)	**34.70%**	58.80%	**4.50%**	**85.29%**

dataset. The results show the effectiveness and superiority of our methods. Moreover, our model tends to predict scene labels while achieving instance segmentation with better performance compared to single task.

1) Comparisons With CNN-Based Scene-Recognition Methods: The existing CNN-based scene-recognition methods achieve their best performance always rely on multiscale fusion. These methods are applied to overcome the shortcoming of the various scales of the local semantic regions. Due to the robust feature extraction capability of the instance segmentation network and the flexible spatial information modeling capability of the GNN, IBMG-Net can effectively utilize multiscale semantic information.

To validate the superiority of our methods, we use three promising classification CNNs as baselines [65], [66], and we compare these works with IBMG-Net on single scale in Table II. The three models, pretrained on ImageNet, are adopted as backbone CNN models in the comparison and fine-tuning on the MCF datasets. We perform a series of experiments to compare the previous methods with our methods. As summarized in Table II, compared to existing CNN methods, IBMG-Net achieves better performance with a clear margin (6.39% over resnet50).

2) Comparisons With Current Instance Segmentation Methods: Unlike the current remote sensing scene-recognition task, the IBMG-Net implements pixel-level prediction of instance components while achieving prediction of scene categories. To further explore the superiority of our IBMG-Net, for instance, segmentation, we perform a series of experiments of classical instance segmentation algorithms on our dataset. The two-step instance segmentation methods, such as mask scoring RCNN, cascade RCNN, and hybrid task cascade (HTC), are most widely used in instance segmentation for remote sensing images. And HTC is the-state-of-the-art method in instance segmentation tasks for natural scenes. Compared with these methods, the IBMG-Net achieves excellent performance on a single-instance segmentation task and is comparable to HTC without adding any data-enhancement operations.

In Table II, the best performances are marked in bold. The classical instance segmentation algorithm improves instance extraction accuracy for most instance categories, and the AP50 metric is significantly improved. However, the classical method is ineffective for target extraction within scenes with large-scale differences, thus hindering the development of mean average accuracy (mAP). In contrast, IBMG-Net utilizes MS-ROI and SCM to alleviate huge scale difference, outperforming all of the classical algorithms in the metric of mAP, and up to 4.50% over baseline (maskRCNN).

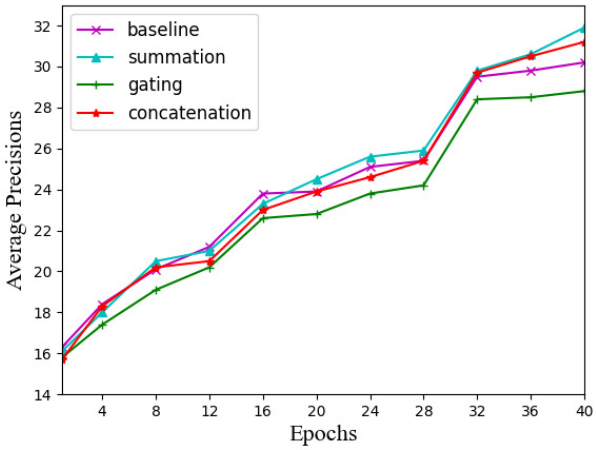


Fig. 10. Ablation analysis on different RoI feature aggregation operation. We visualize the average precision trend on training.

3) Effect of MS-ROI Module: The MSFE mainly contains two key components: MS-ROI and SCM. And the critical operation of the MS-ROI is multiscale feature aggregation. We can use common feature aggregation operations such as concatenation and summation and the gating mechanism to aggregate the extracted region of interest. Fig. 10 shows the comparison between the three choices and a single-layer RoI extractor module (such as our baseline), where the experiments only trained in a single task. And looking at the results of concatenation, we find that the aggregation of different layers of FPN is not always helpful, which is the same as the argument in [59]. Combine the results of the three operations, and we selected the summation as the aggregation of our MS-ROI.

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TABLE III
ABLATION EXPERIMENTS OF MS-ROI MODULE AND SCM MODULE IN MSFE

	MS-ROI	SCM	mAP	$AP50$	AP_S	AP_M	AP_L	Acc
PANet [68]	–	–	31.10%	57.40%	39.30%	34.90%	36.00%	82.33%
GCNet [54]	–	–	31.40%	54.30%	37.30%	37.20%	37.60%	83.01%
mask RCNN(baseline) [40]	–	–	30.20%	53.80%	24.80%	34.20%	33.20%	79.42%
	✓	–	31.60%	56.10%	37.50%	36.30%	37.90%	83.32%
	–	✓	32.20%	56.40%	38.10%	37.50%	38.40%	81.89%
	✓	✓	**33.50%**	**58.30%**	**42.00%**	**40.10%**	**39.70%**	**84.41%**

To further explore the effect of the MSFE module, we construct the multitask framework with the MSFE module and IBGN and achieve better results by joint multitasking training. Then, we append the MSFE module to the basic framework and validate the effect of MS-ROI and SCM by ablation experiments in Table III. We found that the mAP is increased by 1.4% and the Acc is increased by 3.7% with the MS-ROI module, and the improvement of the AP_S and AP_L also verifies that the MS-ROI mitigates the effects of large differences in instance scales on the model. Then we append the SCM module to the framework and improve mAP by 2.0%, MP50 by 2.6% and Acc by 2.47%. When the MSFE uses both the MS-ROI module and SCM module, our model exhibits optimal performance with 3.2%, 4.5%, and 4.99% improvement over the baseline on mAP, AP50, and Acc, respectively.

Meanwhile, we utilize a path aggregation network (PANet) [68] on our dataset and validate the effect of our multiscale region of interest. And our multiscale feature preservation-based approach has an improvement of the APs nearly 2% at multiple scales compared to the APs of PANet.

4) Effect of SCM Module: The SCM samples a sparse set of representative nodes to aggregate global semantic information. As mentioned in Table I, we compare the instance segmentation results of choosing a different number of sample nodes. These results show that all the models based on SCMs can achieve performance improvement over baseline (mask RCNN). Moreover, the instance segmentation task achieves the best performance when M is 9. Therefore, we set 9 as the default in the model.

As summarized in Table III, when we only use the SCM module to aggregate long-range contextual information in a MSFE network, the model has some improvement in the metrics of AP compared to baseline, especially in the metric of AP50. The accuracy improvement for Acc also shows that SCM is robust to discriminative feature representation.

To further validate the superiority of SCM over other attention mechanisms in networks, we used GCNet [54], which is proposed based on nonlocal networks and has a stronger ability to model global information. The results show that GCNet has significantly improved the average accuracy for large-scale targets and medium-scale targets. And the SCM module still has a substantial advantage over GCNet.

5) Effect of IBGN Module: We use an IBGN module to model the spatial relations in complex remote sensing scenes and aggregate instance features to predict scene labels by GC. In the graph network construction stage, to obtain a representation of the spatial relationship between different instances, we have various ways to generate the spatial edge. In addition to using spatial distances directly as edges, we can also use the reciprocal of distances, or the method proposed in [28] that concatenates the position information of two nodes and then use one-layer fully connected network to generate the spatial edge.

TABLE IV
ABLATION ANALYSIS ON DIFFERENT METHODS OF SCENE LABEL PREDICTION

methods	edge	Acc
IBGN	concatenate	82.27%
	distance	**85.29%**
	reciprocal of distance	84.94%
ResNet50	-	78.90%
ResNet50+SCM	-	80.02%

With the experiments based on these three edge-construction strategies, we found that the effect of the method directly based on spatial distance is nearly 3% better on complex facility recognition in remote sensing, while the method directly using spatial distance is 0.35% better than using the reciprocal of distance. We can extrapolate from this, based on spatial distance rather than the reciprocal of distance, allowing us to focus more on key outlier components in addition to the aggregated component pairs.

To further explore the superiority of the IBGN module, we compare IBGN module with two different CNN-based methods of scene recognition. In addition to the resnet50-based method above, the other method uses the output of the backbone (resnet50), followed by the SCM and the global average pooling layer. As summarized in Table IV, our IBGN module performs the accuracy of the CNN-based methods.

The results of Table IV reveal that the IBGN module can achieve the best accuracy of complex facility recognition in the case of limited iteration epochs and limited amount of dataset. And the performance improvement in the second CNN-based method also exposes that the SCM module for aggregating

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Fig. 11. Experiment results of IBMG-Net and classic instance segmentation models on the test set. We visualize the results by assigning different colors to different instances. Unlike the existing methods, the IBMG-Net predicts scene labels while performing instance segmentation results. (a) Ground Truth. (b) MaskRCNN (baseline). (c) Hybrid Task Cascade. (d) IBMG-Net.

TABLE V
INSTANCE SEGMENTATION RESULTS OF BASELINE WITH SINGLE-TASK AND MULTITASK TRAINING

baseline	mAP	$AP50$	AP_S	AP_M	AP_L
Single-task	30.20%	53.80%	24.80%	34.20%	33.20%
Multi-task	32.40%	59.30%	41.40%	38.80%	34.60%

long-range context information is also applicable in the scene recognition task.

6) Effect of Multitask Joint Training: Table V summarizes the instance segmentation results of the baseline under single-task training and multitask joint training, respectively. In the multitask case, we add the IBGN module to the baseline (Mask RCNN) and joint training without adding any data augmentation operations or training strategies. And compared with the single-task training, multitask joint training improves various evaluation metrics, including mAP by 2.2%, AP50 by 5.5%, and AP for instances with different sizes. Moreover, the apparent precision improvement of AP_S (nearly 17%) also attracted our attention. And to reveal the performance trend for different instance categories, we show average category precision (AP) in Table V. The multitask learning is equivalent to adding some prior knowledge, which makes the model more sensitive to the key component in different scenes, especially those with regular shapes and medium scales, such as columnar buildings, court, and storage tanks.

D. Discussion

We visualize the experiment results of IBMG-Net and two current instance segmentation methods in Fig. 11. Specially, we choose three different types of complex geospatial facilities: the petrochemical plant with a dense distribution of component objects, the sewage treatment plant with sparse distribution of component objects, and the nuclear power plant with a regular distribution of components. As a result, our method is significantly better than mask RCNN (baseline), and the performance on large objects is comparable to that of the state-of-the-art (HTC). At the same time, IBMG-Net has a better performance in densely distributed scenes.

To analyze the problems of IBMG-Net, we also visualize three main types of errors in Fig. 12. The experimental errors list is as follows: 1) the recognition of the edges of component objects is insufficient; 2) difficulty in recognizing objects with complex texture appearance; and 3) incorrect prediction of the type of complex facilities, e.g., the fossil-fuel power plant incorrectly predicted as the petrochemical plant.

There are three main reasons for these errors. First, the IBMG-Net focuses on the scale and semantic of the

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Fig. 12. Three main types of errors on the test set. (a) Poor edge recognition in large-scale components. (b) Prediction loss exists for a few spacially shaped component objects. (c) Misjudged complex facility type, e.g., the fossil-fuel power plant incorrectly predicted as the petrochemical plant.

components but ignores the improvement of edge recognition. Second, due to insufficient dataset scale and category imbalance of training samples, some prediction loss occurs for some component objects with complex texture appearance. At the same time, error accumulation of the multistage model is another reason since we use a two-stage method in instance segmentation. The errors generated during coordinate regression and mask prediction inevitably pass to the scene recognition task.

In the future, we will try to perform the single-stage instance segmentation method for alleviating the error accumulation of the multistage model. Besides, we also try to improve the boundary of the IBMG-Net in terms of speed and accuracy. Finally, we will continue to expand the MCF dataset to more complex remote sensing scenes and aim to release the final version of the dataset as soon as possible.

V. Conclusion

In this article, we propose IBMG-Net, a novel multitasks framework, for complex facility recognition in remote sensing imagery. The complex geospatial facilities are composed of several component objects. Inspired by the idea of GNN, we construct a novel graph neural network. Compared with conventional methods, we perform instance segmentation for precise component information. Consider the interference caused by variant scale and strong background information: the MS-ROI retains multiscale features and the SCM enable the network to aggregate global context information. Finally, we achieve performance improvement of two tasks by joint training. To verify the effectiveness of our method, we build the MCF dataset for the MTI of complex facility scenes. The results on the MCF dataset reveal that our multitask framework has significant improvement than other CNN-based models for a single task. In the future, we plan to develop GNN module for further precision improvement experiment and publish the MCF dataset soon.

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